
❖ Set theory

- **Definition:** A set is a well-defined collection of distinct objects, considered as an object in its own right. These objects, called elements, can be anything: numbers, letters, or other abstract entities. Sets are typically denoted using curly braces $\{ \}$.

Examples:

- $\{1, 2, 3\}$ is a set containing the numbers 1, 2, and 3.
- $\{a, b, c\}$ is a set containing the letters a, b, and c.
- $\{\text{red, green, blue}\}$ is a set containing the colors red, green, and blue.

- **Method of representation of set:**

Example: The set of vowels in the English alphabet can be represented as $A = \{a, e, i, o, u\}$.

Example: The set of natural numbers less than 5 can be represented as $N = \{1, 2, 3, 4\}$

Example: The set of all even numbers less than 10 can be represented as $A = \{x : x \text{ is an even number and } x < 10\}$.

Example: The set of all natural numbers divisible by 3 can be represented as $B = \{x \mid x \in \mathbb{N} \text{ and } x \text{ is a multiple of } 3\}$

- **Types of Number Sets:**

- **Natural Numbers (N):** These are the counting numbers, starting from 1 (1, 2, 3, ...).
- **Whole Numbers (W):** This set includes all natural numbers and zero (0, 1, 2, 3, ...).
- **Integers (Z):** Integers encompass all whole numbers and their negative counterparts (... , -3, -2, -1, 0, 1, 2, 3, ...).
- **Rational Numbers (Q):** These are numbers that can be expressed as a fraction p/q , where p and q are integers and q is not zero (e.g., $1/2$, $-3/4$, 5).
- **Irrational Numbers (Q'):** These are numbers that cannot be expressed as a fraction (e.g., π , the square root of 2).
- **Real Numbers (R):** This set includes all rational and irrational numbers.

- **Finite Set:** A finite set is a set that has a countable number of elements.

In other words, you can count how many elements it has, and the counting ends. They will be called a finite set.

Eg. $A = \{1, 2, 3, 4\} \rightarrow$ Finite (4 elements)

Eg. $B = \{\text{"apple"}, \text{"banana"}\} \rightarrow$ Finite (2 elements)

Eg. $C = \emptyset$ (Empty set) $\rightarrow$ Also a finite set (0 elements)

- **Infinite Set:** Infinite Sets are those that have an infinite number of elements present, cases in which the number of elements is hard to determine are known as infinite set.

Eg. $A = \{1, 2, 3, 4, 5, 6, \dots\} \rightarrow$ Infinite

Eg. $B =$ Set of all stars in the sky $\rightarrow$ Infinite

Eg. $C =$ Set of all natural numbers $\rightarrow$ Infinite

- **Empty Set:** The sets with no element/elements is called Empty set. They are denoted as ϕ or $\{\}$ also known as ϕ . Empty Set is also known as Null sets or Void sets.

Eg. $A = \{a : a \text{ is a number greater than } 5 \text{ and less than } 3\}$

Eg. $B = \{p : p \text{ are the students studying in class 7 and class 8}\}$

- **Singleton Set:** Singleton Sets are those sets that have only 1 element present in them.

Eg. $A = \{1\}$ is a singleton set as it has only one element, that is, 1.

Eg. $P = \{a : a \text{ is an even prime number}\}$ is a singleton set as it has only one element 2.

- **Subset:** Set A will be called the Subset of Set B if all the elements present in Set A already belong to Set B. The symbol used for the subset is $\subseteq$.

If A is a Subset of B, it will be written as $A \subseteq B$.

Eg. Set A = {33, 66, 99}

Set B = {22, 11, 33, 99, 66}

Then, $A \subseteq B$

- **Proper Subset:** A proper subset is a subset that contains some but not all elements of the original set. It is not equal to the original set. If A and B are sets, and A is a proper subset of B, we write: $A \subset B$.

Eg. Let B = {1, 2, 3}

Proper subsets of B are:

$\emptyset$, {1}, {2}, {3}, {1, 2}, {1, 3}, {2, 3}

{1, 2, 3} is a subset of B, but not a proper subset, because it is equal to B.

- **Power Set:** The set of any set A is defined as the set containing all the subsets of set A. It is denoted by the symbol $P(A)$ and read as the Power set of A.
For any set A containing n elements, the total number of subsets formed is 2^n . Thus, the power set of A, $P(A)$ has 2^n elements

Eg. For any set $A = \{a, b, c\}$, the power set of A is?

Power Set $P(A)$ is,

$P(A) = \{\emptyset, \{a\}, \{b\}, \{c\}, \{a, b\}, \{b, c\}, \{c, a\}, \{a, b, c\}\}$

- **Universal Set:** A universal set is a set that contains all the elements of the rest of the sets. It can be said that all the sets are the subsets of Universal set. The universal set is denoted as U.

Eg: For Set A = {a, b, c, d} and Set B = {1, 2}, find the universal set containing both set
Universal Set U is, $U = \{a, b, c, d, e, 1, 2\}$

- **Complement of a set:** The complement of a set refers to the elements not in the set, but present in the universal set. U is the universal set and A is a subset of U, then the complement of A, written as A' or A^c or $\bar{A}$.

Eg. If $U = \{1, 2, 3, 4, 5, 6, 7\}$ and $A = \{2, 4, 6\}$ then
 $A' = \{1, 3, 5, 7\}$



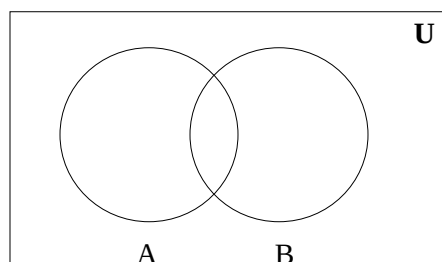
- **Cardinality of sets:** The cardinality of a set is the number of elements in that set and it is usually denoted by $|A|$.

Eg. Let A = {1, 2, 3, 4} then $|A| = 4$

Eg. Let B = {apple, banana, mango} then $|B| = 3$

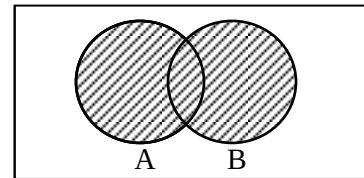
Eg. Let C = {} (empty set) then $|C| = 0$

- **Venn diagram:** A Venn diagram is a visual representation used in set theory to show the relationships between different sets — such as intersection, union, complement, and difference. It uses overlapping circles to represent sets, with the universal set usually shown as a rectangle enclosing the circles.



- **Union of two sets:** The union of two sets contains all elements that are present in the first set, the second set, or elements are in both sets.

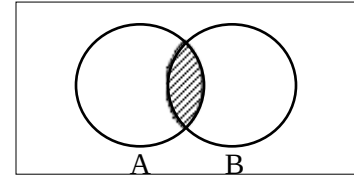
Eg: $A = \{1, 2, 3, 4\}$ and $B = \{3, 5, 6, 7, 9\}$
 $A \cup B = \{1, 2, 3, 4, 5, 6, 7, 9\}$



$(A \cup B)$

- **Intersection of two sets:** The intersection of two sets is a set containing only the elements that are common to both of the original sets.

Eg: $A = \{1, 2, 3, 4\}$ and $B = \{3, 5, 6, 7, 9\}$
 $A \cap B = \{3\}$



$(A \cap B)$

- **Disjoint sets:** Two sets are said to be disjoint when they have no common element. If a collection has two or more sets, the condition of disjointness will be the intersection of the entire collection should be empty.

Eg: $A = \{1, 2, 3, 4\}$ and $B = \{5, 6, 7\}$
 then $A \cap B = \emptyset$

- **Equality of sets:** Two sets are equal if and only if they have the same elements. More formally, for any sets A and B, $A = B$

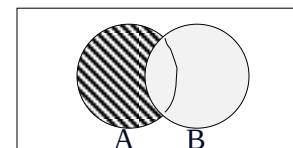
Eg: $A = \{1, 2, 3\}$ and $B = \{2, 1, 3\}$
 $A = B$

- **Equivalent of sets:** Two sets A and B are said to be equivalent if they have the same cardinality number i.e. $n(A) = n(B)$. Generally, we can say that two sets are equivalent to each other if the number of elements in both sets is equal. It is not necessary that both the set have similar elements, or they are a subset of each other

Eg: $A = \{1, 2, 3\}$ and $B = \{a, b, c\}$
 $n(A) = 3 = n(B)$

- **Difference set:** The difference of two sets A and B is defined as the lists of all the elements that are in set A but that are not present in set B. The set notation used to represent the difference between the two sets A and B is $A - B$.

Eg: $A = \{1, 2, 3, 4, 5\}$ and $B = \{3, 4, 5, 6, 7\}$
 $A - B = \{1, 2, 3, 4, 5\} - \{3, 4, 5, 6, 7\} = \{1, 2\}$
 $B - A = \{3, 4, 5, 6, 7\} - \{1, 2, 3, 4, 5\} = \{6, 7\}$



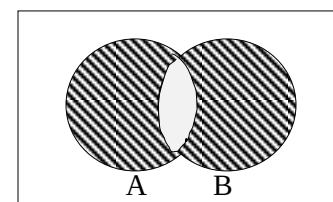
$(A - B)$

- **Symmetric Difference set:** The symmetric difference between two sets A and B is denoted as $A \Delta B$ and is defined as the set of elements that are in either of set but not in their intersection.

Mathematically, this can be expressed as:

$$A \Delta B = (A - B) \cup (B - A) \text{ OR } (A \cup B) - (A \cap B)$$

Eg: $A = \{1, 2, 3, 4, 5\}$ and $B = \{3, 4, 5, 6, 7\}$
 $A - B = \{1, 2, 3, 4, 5\} - \{3, 4, 5, 6, 7\} = \{1, 2\}$
 $B - A = \{3, 4, 5, 6, 7\} - \{1, 2, 3, 4, 5\} = \{6, 7\}$
 $A \Delta B = (A - B) \cup (B - A)$
 $= \{1, 2\} \cup \{6, 7\}$
 $= \{1, 2, 6, 7\}$



$A \Delta B$

- **Cartesian product of set:** The Cartesian product of two sets and , written $A \times B$, is the set of all ordered pairs in which the first element belongs to and the second belongs to : $A \times B = \{ \langle a, b \rangle : a \in A, b \in B \}$. Notice that the size of $A \times B$ is the size of times the size of, that is, $|A \times B| = |A|$

Eg: 1 $A = \{1, 2, 3\}$ and $B = \{4, 5\}$

$$A \times B = \{(1, 4), (1, 5), (2, 4), (2, 5), (3, 4), (3, 5)\}$$

Eg: 2 $A = \{1, 2, 3\}$ and $B = \{a, b, c\}$

$$A \times B = \{(1, a), (1, b), (1, c), (2, a), (2, b), (2, c), (3, a), (3, b), (3, c)\}$$

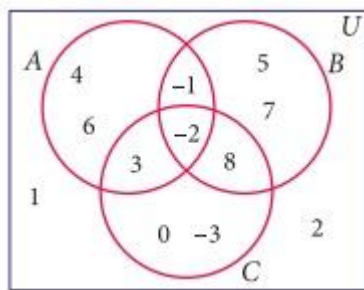
Eg: 3 $A = \{1, 2, 3\}$ and $B = \{4, 5, 6\}$ and $C = \{7, 8\}$

$$A \times B \times C = \{(1, 4, 7), (1, 5, 7), (1, 6, 7), (1, 4, 8), (1, 5, 8), (1, 6, 8), (2, 4, 7), (2, 5, 7), (2, 6, 7), (2, 4, 8), (2, 5, 8), (2, 6, 8), (3, 4, 7), (3, 5, 7), (3, 6, 7), (3, 4, 8), (3, 5, 8), (3, 6, 8)\}$$

- **Properties of set operation:**

- $A \cup B = B \cup A$ (Commutative law)
- $A \cap B = B \cap A$ (Commutative law)
- $(A \cup B) \cup C = A \cup (B \cup C)$ (Associative Law)
- $(A \cap B) \cap C = A \cap (B \cap C)$ (Associative law)
- $A \cup \emptyset = A$ (Law of identity element, $\emptyset$ is the identity of $\cup$)
- $\emptyset \cap A = \emptyset, U \cap A = A$ (Law of $\emptyset$ and U)
- $A \cup A = A$ (Idempotent law)
- $A \cap A = A$ (Idempotent law)
- $U \cup A = U$ (Law of U)
- $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ (Distributive law)
- $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ (Distributive law)
- Complement Laws: $A \cup A' = U$ (ii) $A \cap A' = \emptyset$
- De Morgan's Laws: (i). $(A \cup B)' = A' \cap B'$
- De Morgan's Laws: (ii) $(A \cap B)' = A' \cup B'$
- Law of Double Complementation: $(A')' = A$
- Law of Empty Set and Universal Set: $\emptyset' = U$ and $U' = \emptyset$
- If $|A| = m, |B| = n$ then $|A \times B| = |B \times A| = mn$
- $A \times B = B \times A$ iff $A = B; A, B \neq \emptyset$
- $A \times (B \cup C) = (A \times B) \cup (A \times C)$
- $A \times (B \cap C) = (A \times B) \cap (A \times C)$
- $(A \times B) \cap (C \times D) = (A \cap C) \times (B \cap D)$
- $n(A \cup B) = n(A) + n(B) - n(A \cap B)$
- $n(A \cap B) = n(A) + n(B) - n(A \cup B)$
- $n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(B \cap C) - n(A \cap C) + n(A \cap B \cap C)$

1. Using the adjacent Venn diagram, find the following sets:



(i) $A - B$ (ii) $B - C$ (iii) $A' \cup B'$ (iv) $A' \cap B'$ (v) $(B \cup C)'$ (vi) $A - (B \cup C)$ (vii) $A - (B \cap C)$

- (i) $A - B = \{3, 4, 6\}$
(ii) $B - C = \{-1, 5, 7\}$
(iii) $A' \cup B'$
 $A' = \{1, 2, 0, -3, 5, 7, 8\}$
 $B' = \{-3, 0, 1, 2, 3, 4, 6\}$
 $A' \cup B' = \{-3, 0, 1, 2, 3, 4, 5, 6, 7, 8\}$
(iv) $A' \cap B'$
 $A' \cap B' = \{-3, 0, 1, 2\}$
(v) $(B \cup C)'$
 $B \cup C = \{-3, -2, -1, 0, 3, 5, 7, 8\}$
 $(B \cup C)' = U - (B \cup C)$
 $= \{\cancel{3}, \cancel{2}, \cancel{1}, 0, 1, 2, \cancel{3}, 4, \cancel{5}, 6, \cancel{7}, \cancel{8}\} - \{-3, -2, -1, 0, 3, 5, 7, 8\}$
 $(B \cup C)' = \{1, 2, 4, 6\}$
(vi) $A - (B \cup C) = \{\cancel{2}, \cancel{1}, \cancel{3}, 4, 6\} - \{-3, -2, -1, 0, 3, 5, 7, 8\} = \{4, 6\}$
(vii) $A - (B \cap C)$
 $B \cap C = \{-2, 8\}$
 $A - (B \cap C) = \{\cancel{2}, -1, 3, 4, 6\} - \{-2, 8\} = \{-1, 3, 4, 6\}$

2. If $A = \{x : x \in \mathbb{Z}, -2 < x \leq 4\}$, $B = \{x : x \in \mathbb{W}, x \leq 5\}$, $C = \{-4, -1, 0, 2, 3, 4\}$, then verify $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$.

$$\begin{aligned} A &= \{x : x \in \mathbb{Z}, -2 < x \leq 4\} = \{-1, 0, 1, 2, 3, 4\} \\ B &= \{x : x \in \mathbb{W}, x \leq 5\} = \{0, 1, 2, 3, 4, 5\} \\ C &= \{-4, -1, 0, 2, 3, 4\} \end{aligned}$$

$$\begin{aligned} A \cup (B \cap C) &= \{-1, 0, 1, 2, 3, 4\} \cup \{0, 2, 3, 4\} = \{-1, 0, 1, 2, 3, 4\} \quad \dots (1) \\ (A \cap B) \cup (A \cap C) &= \{0, 1, 2, 3, 4\} \cup \{-1, 0, 2, 3, 4\} = \{-1, 0, 1, 2, 3, 4\} \quad \dots (2) \end{aligned}$$

From (1) and (2), it is verified that

$$A \cup (B \cap C) = (A \cap B) \cup (A \cap C)$$

3. If $K = \{a, b, d, e, f\}$, $L = \{b, c, d, g\}$ and $M = \{a, b, c, d, h\}$, then find the following:

(i) $K \cup (L \cap M)$ (ii) $K \cap (L \cup M)$ (iii) $(K \cup L) \cap (K \cup M)$ (iv) $(K \cap L) \cup (K \cap M)$

and verify distributive laws.

$$K = \{a, b, d, e, f\}, L = \{b, c, d, g\} \text{ and } M = \{a, b, c, d, h\}$$

(i) $K \cup (L \cap M)$

$$L \cap M = \{b, c, d, g\} \cap \{a, b, c, d, h\} = \{b, c, d\}$$

$$K \cup (L \cap M) = \{a, b, d, e, f\} \cup \{b, c, d\} = \{a, b, c, d, e, f\}$$

(ii) $K \cap (L \cup M)$

$$L \cup M = \{a, b, c, d, g, h\}$$

$$K \cap (L \cup M) = \{a, b, d, e, f\} \cap \{a, b, c, d, g, h\} = \{a, b, d\}$$

(iii) $(K \cup L) \cap (K \cup M)$

$$K \cup L = \{a, b, c, d, e, f, g\}$$

$$K \cup M = \{a, b, c, d, e, f, h\}$$

$$(K \cup L) \cap (K \cup M) = \{a, b, c, d, e, f\}$$

(iv) $(K \cap L) \cup (K \cap M)$

$$(K \cap L) = \{b, d\}$$

$$(K \cap M) = \{a, b, d\}$$

$$(K \cap L) \cup (K \cap M) = \{b, d\} \cup \{a, b, d\} = \{a, b, d\}$$

Distributive laws

$$K \cup (L \cap M) = (K \cup L) \cap (K \cup M)$$

$$\{a, b, c, d, e, f\} = \{a, b, c, d, e, f, g\} \cap \{a, b, c, d, e, f, h\} = \{a, b, c, d, e, f\}$$

Thus Verified.

$$K \cap (L \cup M) = (K \cap L) \cup (K \cap M)$$

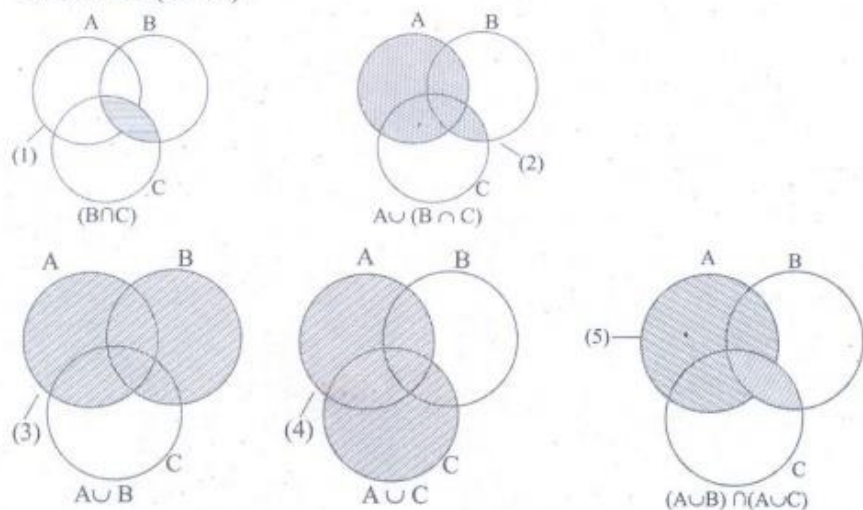
$$\{a, b, d\} = \{a, b, c, d, e, f, g\} \cup \{a, b, c, d, e, f, h\}$$

$$= \{a, b, d\}$$

Thus Verified.

4. Verify $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ using Venn diagrams.

L.H.S. $A \cup (B \cap C)$



From (2) and (5), it is verified that $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

5. If $A = \{b, c, e, g, h\}$, $B = \{a, c, d, g, i\}$ and $C = \{a, d, e, g, h\}$, then show that $A - (B \cap C) = (A - B) \cup (A - C)$.

$$\begin{aligned}
 A &= \{b, c, e, g, h\} \\
 B &= \{a, c, d, g, i\} \\
 C &= \{a, d, e, g, h\} \\
 B \cap C &= \{a, d, g\} \\
 A - (B \cap C) &= \{b, c, e, g, h\} - \{a, d, g\} = \{b, c, e, h\} \quad \dots (1) \\
 A - B &= \{b, c, e, g, h\} - \{a, c, d, g, i\} = \{b, e, h\} \\
 A - C &= \{b, c, e, g, h\} - \{a, d, e, g, h\} = \{b, c\} \\
 (A - B) \cup (A - C) &= \{b, c, e, h\} \quad \dots (2)
 \end{aligned}$$

From (1) and (2) it is verified that

$$A - (B \cap C) = (A - B) \cup (A - C)$$

6. If $A = \{-2, 0, 1, 3, 5\}$, $B = \{-1, 0, 2, 5, 6\}$ and $C = \{-1, 2, 5, 6, 7\}$, then show that $A - (B \cup C) = (A - B) \cap (A - C)$.

$$\begin{aligned}
 A &= \{-2, 0, 1, 3, 5\}, B = \{-1, 0, 2, 5, 6\} \\
 C &= \{-1, 2, 5, 6, 7\} \\
 B \cup C &= \{-1, 0, 2, 5, 6, 7\} \\
 A - (B \cup C) &= \{-2, 1, 3\} \quad \dots (1) \\
 (A - B) &= \{-2, 1, 3\} \\
 (A - C) &= \{-2, 0, 1, 3\} \\
 (A - B) \cap (A - C) &= \{-2, 1, 3\} \quad \dots (2)
 \end{aligned}$$

From (1) and (2), it is verified that

$$A - (B \cup C) = (A - B) \cap (A - C)$$

7. If $U = \{4,7,8,10,11,12,15,16\}$, $A=\{7,8,11,12\}$ and $B = \{4,8,12,15\}$ then verify De Morgan's Laws for complementation.

$$U = \{4, 7, 8, 10, 11, 12, 15, 16\}$$

$$A = \{7, 8, 11, 12\}, B = \{4, 8, 12, 15\}$$

De Morgan's Laws for complementation.

$$(A \cup B)' = A' \cap B'$$

$$A \cup B = \{4, 7, 8, 11, 12, 15\}$$

$$(A \cup B)' = \{4, 7, 8, 10, 11, 12, 15, 16\} - \{4, 7, 8, 11, 12, 15\}$$

$$= \{10, 16\} \quad \dots (1)$$

$$A' = \{4, 10, 15, 16\}$$

$$B' = \{7, 10, 11, 16\}$$

$$A' \cap B' = \{10, 16\} \quad \dots (2)$$

From (1) and (2) it is verified that

$$(A \cup B)' = A' \cap B'.$$

8. From the Venn diagram, verify that $n(A \cup B) = n(A) + n(B) - n(A \cap B)$

Solution From the venn diagram,

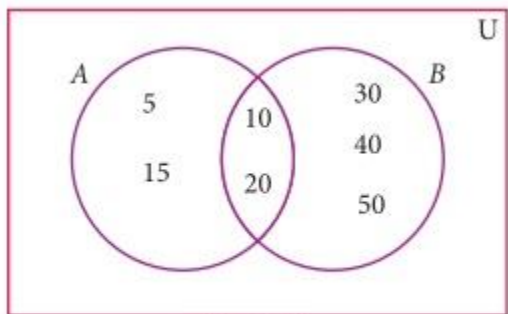


Fig. 1.33

$$A = \{5, 10, 15, 20\}$$

$$B = \{10, 20, 30, 40, 50\}$$

Then

$$A \cup B = \{5, 10, 15, 20, 30, 40, 50\}$$

$$A \cap B = \{10, 20\}$$

$$n(A) = 4, n(B) = 5, n(A \cup B) = 7, n(A \cap B) = 2$$

$$n(A \cup B) = 7 \quad \dots\dots\dots(1)$$

$$n(A) + n(B) - n(A \cap B) = 4 + 5 - 2 = 7 \quad \dots\dots\dots(2)$$

From (1) and (2), $n(A \cup B) = n(A) + n(B) - n(A \cap B)$.

9. Let $A = \{b, d, e, g, h\}$ and $B = \{a, e, c, h\}$. Verify that $n(A-B) = n(A) - n(A \cap B)$.

Solution

$$A = \{b, d, e, g, h\}, B = \{a, e, c, h\}$$

$$A - B = \{b, d, g\}$$

$$n(A - B) = 3 \quad \dots (1)$$

$$A \cap B = \{e, h\}$$

$$n(A \cap B) = 2, n(A) = 5$$

$$n(A) - n(A \cap B) = 5 - 2 = 3 \quad \dots (2)$$

From (1) and (2) we get $n(A - B) = n(A) - n(A \cap B)$.

10. In a school, all students play either Hockey or Cricket or both. 300 play Hockey, 250 play Cricket and 110 play both games. Find

(i) The number of students who play only Hockey.

(ii) The number of students who play only Cricket.

(iii) The total number of students in the School.

Engineering education courses

Solution:

Let H be the set of all students who play Hockey and C be the set of all students who play Cricket.

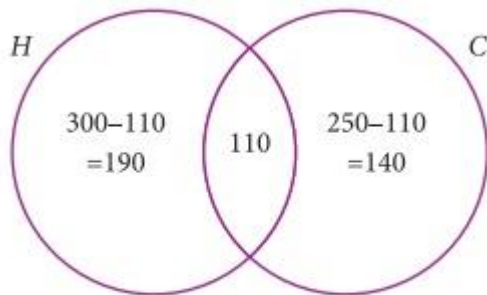


Fig. 1.34

Then $n(H) = 300$, $n(C) = 250$ and $n(H \cap C) = 110$

Using Venn diagram,

From the Venn diagram,

(i) The number of students who play only Hockey = 190

(ii) The number of students who play only Cricket = 140

(iii) The total number of students in the school = $190+110+140 = 440$

Aliter

(i) The number of students who play only Hockey

$$n(H-C) = n(H) - n(H \cap C)$$

$$= 300 - 110 = 190$$

(ii) The number of students who play only Cricket

$$n(C-H) = n(C) - n(H \cap C)$$

$$= 250 - 110 = 140$$

(iii) The total number of students in the school

$$n(H \cup C) = n(H) + n(C) - n(H \cap C)$$

$$= 300 + 250 - 110 = 440$$

11. In a party of 60 people, 35 had Vanilla ice cream, 30 had Chocolate ice cream. All the people had at least one ice cream. Then how many of them had,

(i) Both Vanilla and Chocolate ice cream.

(ii) Only Vanilla ice cream.

(iii) Only Chocolate ice cream.

Solution:

Let V be the set of people who had Vanilla ice cream and C be the set of people who had Chocolate ice cream.

Then $n(V) = 35$, $n(C) = 30$, $n(V \cup C) = 60$,

Let x be the number of people who had both ice creams.

From the Venn diagram

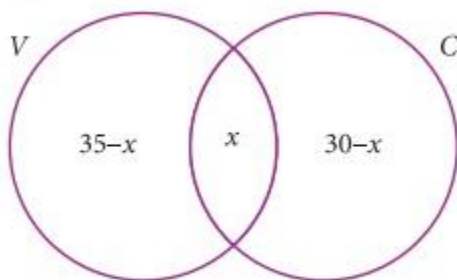


Fig. 1.35

$$35 - x + x + 30 - x = 60$$

$$65 - x = 60$$

$$x = 5$$

Hence 5 people had both ice creams.

(i) Number of people who had only Vanilla ice cream = $35 - x$

$$= 35 - 5 = 30$$

(ii) Number of people who had only Chocolate ice cream = $30 - x$

$$= 30 - 5 = 25$$

We have learnt to solve problems involving two sets using the formula $n(A \cup B) = n(A) + n(B) - n(A \cap B)$. Suppose we have three sets, we can apply this formula to get a similar formula for three sets.

For any three finite sets A , B and C

$$n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(B \cap C) - n(A \cap C) + n(A \cap B \cap C)$$

12. In a college, 240 students play cricket, 180 students play football, 164 students play hockey, 42 play both cricket and football, 38 play both football and hockey, 40 play both cricket and hockey and 16 play all the three games. If each student participate in at least one game, then find (i) the number of students in the college (ii) the number of students who play only one game.

Solution:

Let C , F and H represent sets of students who play Cricket, Football and Hockey respectively.

Then, $n(C) = 240$, $n(F) = 180$, $n(H) = 164$, $n(C \cap F) = 42$,

$n(F \cap H) = 38$, $n(C \cap H) = 40$, $n(C \cap F \cap H) = 16$.

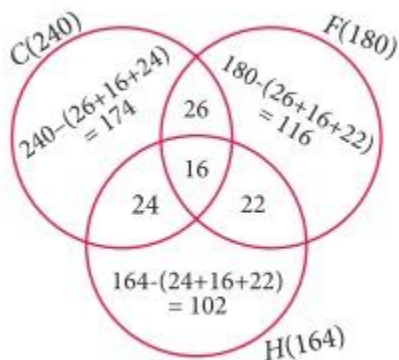


Fig.1.37

Let us represent the given data in a Venn diagram.

(i) The number of students in the college

$$= 174 + 26 + 116 + 22 + 102 + 24 + 16 = 480$$

(ii) The number of students who play only one game

$$= 174 + 116 + 102 = 392$$

13. In a residential area with 600 families $\frac{3}{5}$ owned scooter, $\frac{1}{3}$ owned car, $\frac{1}{4}$ owned bicycle, 120 families owned scooter and car, 86 owned car and bicycle while 90 families owned scooter and bicycle. If $\frac{2}{15}$ of families owned all the three types of vehicles, then find (i) the number of families owned at least two types of vehicle. (ii) the number of families owned no vehicle.

Solution: Let S , C and B represent sets of families who owned Scooter, Car and Bicycle respectively.

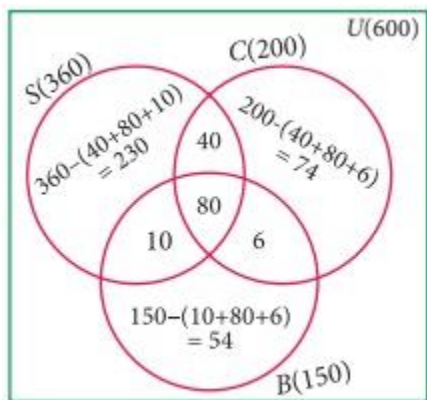


Fig.1.38

Given, $n(U) = 600$

$$n(S) = \frac{3}{5} \times 600 = 360$$

$$n(C) = \frac{1}{3} \times 600 = 200$$

$$n(B) = \frac{1}{4} \times 600 = 150$$

$$n(S \cap C \cap B) = \frac{2}{15} \times 600 = 80$$

From Venn diagram,

(i) The number of families owned at least two types of vehicles = $40 + 6 + 10 + 80 = 136$

(ii) The number of families owned no vehicle

$$= 600 - (\text{owned at least one vehicle})$$

$$= 600 - (230 + 40 + 74 + 6 + 54 + 10 + 80)$$

$$= 600 - 494 = 106$$

14. In a group of 100 students, 85 students speak Tamil, 40 students speak English, 20 students speak French, 32 speak Tamil and English, 13 speak English and French and 10 speak Tamil and French. If each student knows atleast any one of these languages, then find the number of students who speak all these three languages.

Solution: Let A , B and C represent sets of students who speak Tamil, English and French respectively.

$$\text{Given, } n(A \cup B \cup C) = 100, n(A) = 85, n(B) = 40, n(C) = 20,$$

$$n(A \cap B) = 32, n(B \cap C) = 13, n(A \cap C) = 10.$$

We know that,

$$n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(B \cap C) - n(A \cap C) + n(A \cap B \cap C)$$

$$100 = 85 + 40 + 20 - 32 - 13 - 10 + n(A \cap B \cap C)$$

$$\text{Then, } n(A \cap B \cap C) = 100 - 90 = 10$$

Therefore, 10 students speak all the three languages.

15. (i) If $n(A) = 25$, $n(B) = 40$, $n(A \cup B) = 50$ and $n(B') = 25$, find $n(A \cap B)$ and $n(U)$.

(ii) If $n(A) = 300$, $n(A \cup B) = 500$, $n(A \cap B) = 50$ and $n(B') = 350$, find $n(B)$ and $n(U)$.

$$\begin{aligned} \text{(i)} \quad n(A \cap B) &= n(A) + n(B) - n(A \cup B) \\ n(A \cap B) &= 25 + 40 - 50 = 65 - 50 = 15 \\ n(U) &= n(B) + n(B') = 40 + 25 = 65 \end{aligned}$$

$$\begin{aligned} \text{(ii)} \quad n(U) &= n(B) + n(B') \\ n(A \cap B) &= n(A) + n(B) - n(A \cup B) \\ n(B) &= n(A \cup B) + n(A \cap B) - n(A) = 500 + 50 - 300 = 250 \\ n(U) &= 250 + 350 = 600. \end{aligned}$$

16. (i) If $n(A) = 25$, $n(B) = 40$, $n(A \cup B) = 50$ and $n(B') = 25$, find $n(A \cap B)$ and $n(U)$.

(ii) If $n(A) = 300$, $n(A \cup B) = 500$, $n(A \cap B) = 50$ and $n(B') = 350$, find $n(B)$ and $n(U)$.

$$\begin{aligned} \text{(i)} \quad n(A \cap B) &= n(A) + n(B) - n(A \cup B) \\ n(A \cap B) &= 25 + 40 - 50 = 65 - 50 = 15 \\ n(U) &= n(B) + n(B') = 40 + 25 = 65 \end{aligned}$$

$$\begin{aligned} \text{(ii)} \quad n(U) &= n(B) + n(B') \\ n(A \cap B) &= n(A) + n(B) - n(A \cup B) \\ n(B) &= n(A \cup B) + n(A \cap B) - n(A) = 500 + 50 - 300 = 250 \\ n(U) &= 250 + 350 = 600. \end{aligned}$$

17. Verify $n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(B \cap C) - n(A \cap C) + n(A \cap B \cap C)$ for the following sets.

(i) $A = \{a, c, e, f, h\}$, $B = \{c, d, e, f\}$ and $C = \{a, b, c, f\}$

(ii) $A = \{1, 3, 5\}$, $B = \{2, 3, 5, 6\}$ and $C = \{1, 5, 6, 7\}$

$$n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(B \cap C) - n(A \cap C) + n(A \cap B \cap C)$$

$$(i) \quad A = \{a, c, e, f, h\}, B = \{c, d, e, f\}, C = \{a, b, c, f\}$$

$$n(A) = 5, n(B) = 4, n(C) = 4$$

$$n(A \cap B) = 3$$

$$n(B \cap C) = 2$$

$$n(A \cap C) = 3$$

$$n(A \cap B \cap C) = 2$$

$$A \cap B = \{c, e, f\}$$

$$B \cap C = \{c, f\}$$

$$A \cap C = \{a, c, f\}$$

$$A \cap B \cap C = \{c, f\}$$

$$A \cup B \cup C = \{a, c, d, e, f, b, h\}$$

$$\therefore n(A \cup B \cup C) = 7 \quad \dots (1)$$

$$n(A) + n(B) + n(C) - n(A \cap B) - n(B \cap C) - n(A \cap C) + n(A \cap B \cap C)$$

$$= 5 + 4 + 4 - 3 - 2 - 3 + 2 = 15 - 8 = 7$$

$$\dots (2)$$

$$(1) = (2)$$

$$\Rightarrow n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(B \cap C) - n(A \cap C) + n(A \cap B \cap C)$$

Hence it is verified.

$$(ii) \quad A = \{1, 3, 5\}, B = \{2, 3, 5, 6\}, C = \{1, 5, 6, 7\}$$

$$n(A) = 3, n(B) = 4, n(C) = 4$$

$$n(A \cap B) = 2$$

$$n(B \cap C) = 2$$

$$n(C \cap A) = 2$$

$$n(A \cap B \cap C) = 1$$

$$n(A \cup B \cup C) = 6$$

$$n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(B \cap C) - n(A \cap C) + n(A \cap B \cap C)$$

$$6 = 3 + 4 + 4 - 2 - 2 - 2 + 1 = 12 - 6 = 6$$

Hence it is verified.